

National College of Ireland

Project Submission Sheet

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Date: 24/04/2026.....

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AI Acknowledgement Supplement

Business Intelligence & Business Analytics (MSCDAD_A)

Data Driven Renewable Customer Ecosystem for Greenplus Energy Ireland : A CRM and BI Transformation Techniques

Your Name/Student Number	Course	Date
Ruchita Raut-x24240702 / Divesh Patil -x23401478	Business Intelligence & Business Analytics (MSCDAD_A)	24/04/2026

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AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
Chatgpt	It was used or better understanding of complex Topics	Chatgpt.com

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This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used.**

[Insert Tool Name]	
ChatGPT	
explain me about how the salesforces work in simple language and how do we build any automation tool as simply it and tell me in pragraph	Think of Salesforce CRM as a smart digital system that helps businesses keep track of all their customers and sales activities in one place. Instead of using notebooks or scattered spreadsheets, everything is organized new people are stored as leads,

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Additional Evidence:

explain me about how the salesforces work in simple language and how do we build any automation tool as simply it and tell me in pragraph

Think of Salesforce CRM as a smart digital system that helps businesses keep track of all their customers and sales activities in one place. Instead of using notebooks or scattered spreadsheets, everything is organized—new people are stored as leads, real customers as contacts, and possible deals as opportunities. Salespeople can see who they need to follow up with, what stage a deal is in, and what actions to take next, so nothing slips through the cracks. It basically removes guesswork and keeps the whole sales process structured and visible.

Building automation in it—or in any tool—is actually very simple if you stop making it complicated. Every automation follows one basic idea: “when something happens, do something automatically.” First, you decide the trigger (like a new lead being added), then set conditions (for example, only if the lead is from a certain country), and finally define the action (such as assigning it to a salesperson or sending an email). Tools like Salesforce Flow let you create this visually without coding, like drawing a flowchart. If you can clearly describe the trigger, condition, and action in plain language, you already understand automation—the tool is just there to execute your thinking.

Data Driven Renewable Customer Ecosystem for Greenplus Energy Ireland : A CRM and BI Transformation Techniques

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Abstract— The project aims to solve the various operational and management issues currently experienced by the “GreenPlus Energy Ireland” which is a renewable energy company. The issues include high customer churn, increasing billing disputes and the difficulty in efficiently delivering the services to customers. An integrated CRM and BI solution will be implemented to help relieve these issues. The implementation of Salesforce as a centralised CRM system which will allow for the aggregation of customer data, to streamline delivery of services through an automated service screw, and to create the real time data analytics for decision making through the use of Power BI. The implementation also aims to increase the customer retention and accurate billing, while decreasing the amount of time needed to resolve the customer complaints. Also, the system will allow the development of a data driven marketing approach and improve the relationships with renewable energy partners through the data sharing. Though, this project will show the effectiveness of the CRM and BI integration in improving the business performance and decision making of the energy industry.

Keywords— CRM, Business Intelligence, Salesforce, Power BI, Customer Retention, Renewable Energy, Data Analytics

YOUTUBE VIDEO LINK : <https://youtu.be/16YSxubG-pw>

PART 1 : SPECIFICATION REPORT

1. INTRODUCTION

The Company Name "Greenplus Energy Ireland" which is a two-year-old renewable electricity provider in the Irish market, which is supplying the wind and solar-generated power to homes and small businesses. As in terms of Customers the company in the start gain a strong number of customers ,but now the company is struggling with extend of internal issues which are putting the growth of company at risk. The Customers are leaving at an increasing rate, the billing disputes are going unresolved, the complaints are taking too long to handle, and the marketing efforts are falling flat largely because there is no any central system in place to manage the customer relationships or to track what is actually happening over the business. On top of that, management has no real-time visibility into how the company is performing, which means decisions are being made reactively rather than being driven by data. This project brings in a Business Intelligence and Analytics consultant to take the stock of where things currently stand, identify the gaps, and design a practical solution. where the solution centres on implementing Salesforce as a CRM plays a important role in modern organization by allowing business to manage customer interactions,improve customer satisfaction and enhance the retention rates [1]. CRM integrates customer data over the departments, allowing organizations to make more informed, data-driven decisions[1] and the Power BI to give management clear, and dashboard based visibility into both day-to-day operations and the longer-term performance. The project also looks beyond fixing current problems where it introduces a progressive layer where certain customers could be identified as potential renewable energy partners, and opening up a new avenue for business growth next to the operational improvements.

2. BACKGROUND OF THE ORGANISATION AND THE MARKETPLACE:

A. Company Overview:

The "GreenPlus Energy Ireland" is a young renewable electricity company that sells the green energy packages to the homes and the small businesses over the Ireland. As it is being in working for two years, the company has moved past it's initial launch phase, but haven't yet built that kind of set the internal systems like mature data, customer service, and management system and processes that a growing business needs to work smoothly. Its early growth came from the getting customers through the efforts and the attract of renewable energy offerings, but sadly, the company's internal operations hasn't keep the speed with the growth. Although, the company's team is still depending on the spreadsheets, emails, and the manual tracking, to manage everything from the customer records and the billing queries and from complaints to the marketing outreach. So after all after this messy approach the quality of the service is becoming increasingly inconsistent, where the things are falling, and management simply doesn't have a clear or timely picture of how the business is actually performing day to day.

B. Description of the Product and the Service:

The company "Greenplus Energy Ireland" which is an electricity provider, as they do not sell the products like solar panels. The electricity Greenplus supplies comes from the renewable sources such as wind and the solar. So the customers are essentially getting the same everyday service, as they would get from any other energy provider, but with some added benefits of knowing that there power is coming from clean, and green generation rather than the fossil fuels. In sectors such as energy utilities, the integration of the CRM systems with digital platforms allows real-time data processing and reduces the manual errors, leading to improved customer experience and operational performance [2].

C. Renewable Energy Thought:

Renewable Energy concept is that which refers to the electricity generated from the natural resources such as wind and the sunlight those are sustainable and do not reduce over time. As these sources reduce the environmental impact and the carbon emissions. Although the Greenplus focuses on the supplying the electricity developed from the renewable energy generation.

D. Source of Electricity:

The wind farms and the solar farms which are the renewable energy producers from where the company sources the electricity. The renewable electricity which is distributed through the national Grid that is the country wide network which is responsible for transferring the electricity from the generation sources to the end consumers. Whereas the Greenplus purchases the electricity from this System and it supplies to the customers of it.

E. Market Background:

The Irish energy market is a tough to operate in where the customers have high expectations, specially when it comes to understanding their bills, getting issues to get solved quickly, and seeing a genuine commitment to sustainability. The Renewable energy providers like Greenplus aren't just competing on the price but they also need to earn and maintain the customer trust, the deliver where it should have the quality experience, and back up their green credentials with the reliable service. In that kind of environment, any cracks in the work can very quickly lead to customers walking away and the company's reputation taking a hit. so for a business as young as Greenplus, not having proper integrated systems in place is a real risk ,specially when something as basic as getting a bill right or responding to a complaint can be the difference between a loyal customer and a lost one.

F. Current Business Position:

The company "Greenplus" is currently at a critical point in its journey. Although it has managed to establish itself in the market, but the problems it is facing internally are serious though that they could threaten the company's future, if nothing is done about them. so the real issue here is not just about growing the customer base ,but infact that the company doesn't yet have the right base in place to support the growth of the company. As there's no proper data infrastructure, the processes are still largely manual, and the management doesn't have the information, so it needs to make smart decisions. So what "Greenplus" company needs right now is a dual proposal which is getting its current work under the control and stabilized, while at the same time putting a valid data strategy in place, that sets the company up for sustainable success going forward.

3. BUSINESS PROBLEM STATEMENT

A. BREAKING CUSTOMER AND OPERATIONAL DATA:

The company "Greenplus" has no any single place where all customers and the working information is stored and managed. On the other hand, there data about the customer interactions, the contracts, the billing issues, and the support history is disconnect over emails, spreadsheets, and the various manual files that don't talk to each other. This makes it very difficult to get a clear and the complete picture of any given customer or the situation, and it naturally leads to duplicated records, missing information, and the customers receiving inconsistent service depending on who they speak to or when.

B. Inefficient Complaint Handling and the Billing Arguments:

Without any structured workflows the complaints of Customers are handles manually. Customer issues are managed responsively. There are repeating Billing inaccuracies and the unclear charges which results in arguments. As these issues directly supports to the customer frustration and the increased churn.

C. Weak Marketing and Lack of Management Visibility:

The marketing campaigns are not data-driven and where there is lack of proper sections and the performance tracking of the customers. As the result, the customers engagement rates are low and the customer maintenance efforts are ineffective. The Management does not have the access to the real-time dashboards or the performance metrics. Therefore, the Decision-making is sensible which is usually based on the assumptions rather than data.

D. The Absence of Strategic Growth Model:

The Company currently see that every customer simply as someone who buys electricity and that's where the relationship ends. As there is no process in place to allow some of those customers might actually have the potential to generate there own renewable energy. For example through like rooftop solar panels. This clearly means that company is missing out on an important strategic opportunity, as identifying and engaging those customers which could open up to the business and which will support the long-term growth well beyond just selling electricity.

4. BUSINESS GOALS AND PROJECT OBJECTIVES:

A. Business Goals where it tell what company wants:

1. To Reduce the Customer churn
2. To Reduce the billing arguments
3. To Improve complaint resolution time
4. To Improve campaign engagement and the targeting
5. To Improve management visibility through dashboards
6. To Create a renewable partner lead pipeline for the future growth.

B. Project Objectives where it tells what your solution will do :

1. To Design and implement a CRM System to centralize customer data and service data.
2. To automate the workflows for billing alerts, complaints, and feedback
3. To Developed the Power BI dashboards for business insights
4. To create the data-driven model to identify renewable partner candidates
5. To create a mock but realistic dataset for the implementation and evaluation
6. To encourage measurable improvements in business performance.

5. Stakeholder Analysis:

Table to be Created:

Stakeholder	Role	Interest	Impact
Customers	End users	Service Quality	High
Support Team	Handle Complaints	Efficiency	High
Marketing Team	Campagins	Targeting	Medium
Management	Decision-making	KPIs	Very High
IT Team	System Implementation	Maintenance	Medium

Table 1: Stakeholder Analysis

The stakeholder Analysis solution addresses both of the operational users as well as the decision- makers by the combining the CRM functionality with the analytical dashboards.

6.SWOT Analysis:

In order for the "Greenplus Energy Ireland" to have an accurate understanding of the current state of the company, it conducted a SWOT analysis to capture the conditions of both the in-house business activity as well as the market activity of the company. The analysis was divided into four separate sections such as Strengths, Weaknesses, Opportunities, and the Threats which gives Greenplus Energy Ireland a complete picture of where they are today in relation to these four sections and how the result affects their future.

A. SWOT:

Strengths:

- 1.Green Plus has built a solid reputation as a genuine green energy provider in the Irish market.
- 2.Being an young company, the Greenplus has a efficient and flexible structure which allows it to adapt and make changes quickly.
- 3.The company "Greenplus" has made an promising start, successfully attracting customers and gaining a support in the Irish energy market since launching.
4. There is real scope to engage the customers in more meaningful and creative ways than the company is currently doing.

Weakness:

1. The company Greenplus has no single system in place to manage and track the customer relationships.
2. The company still depends on the manual, and the disconnected processes that make it difficult to manage day-to-day operations efficiently.
3. The Customers are regularly experiencing problems with incorrect bills and the disputes that are taking too long to resolve.
4. The business has no clear way of monitoring or keeping the track of the service it is delivering to the customers.
5. The Marketing efforts are not being targeted effectively, and also there is no proper way to measure whether they are actually working.
6. The company has no any formal process in place to identify or engage customers who could become renewable energy partners.

Opportunities:

1. More and more people are showing an interest in generating there own renewable energy rather than simply consuming it.

2. More and more customers are adopting smart meters in their homes and businesses.
3. Using the customer data to better understand, why people leave and taking action to keep them.
4. Marketing can be made more effective by using the customer data to better target the right people with the right message.
5. Building a network of customers who can also contribute to the renewable energy generation.

Threats:

1. The Customers are leaving at a rate that is becoming difficult to ignore.
2. The Irish energy market is crowded, and Greenplus faces constant pressure from more established competitors with greater resources and stronger brand recognition.
3. Poor service experiences are already beginning to erode the trust and reputation that Greenplus has worked hard to build.
4. Ireland's energy market is heavily controlled, and the customers expect a high standard of service.
5. The Customers in the Irish energy market are not afraid to switch the providers if they feel that they are not getting value or good services.

B.SWOT Analysis Interpretation:

The SWOT analysis shows that Greenplus Company is actually in a decent starting position and its focus on green energy, the early customer growth, and the ability to adapt as a young company, as all give it a solid foundation to build on. But the internal weaknesses are hard to ignore. Without a centralised CRM system, customer data is all over the place, the complaints and the billing issues are being handled manually, and the management has no real visibility into what's going wrong or where. And On top of that, the company has no strategy for identifying customers who could become renewable energy partners, which is a missed opportunity for the future growth.

Where as, looking outward, the timing could actually work in Greenplus's favour. Growing public interest in renewable energy and smart technology creates real opportunities for the company to engage customers in new ways and build a renewable partner ecosystem. However, the threats are equally real that the, customers won't hesitate to switch providers if the service isn't up to scratch, and the regulatory pressure around the transparency and the service standards is only increasing. The overall picture is clear, the company Greenplus has genuine potential, but it needs to sort out its operational foundations now before the weaknesses cause any more damage. This is exactly why a CRM and Business Intelligence solution is needed.

7. Gap Analysis:

A gap analysis is usually about identifying the distance between where the Greenplus company is right now and where it needs to be. While the SWOT analysis shows the company's strategic position, and the gap analysis goes a step further by finding the specific problem areas that are holding the business back from reaching its goals. After understanding these gaps in detail is what lays the groundwork for the CRM and Business Intelligence solution which is being proposed, and making sure that what gets built actually addresses the real issues the company is facing. The structured analytical approach such as CRISP-DM framework highlights the importance of clearly identifying business problems, understanding data gaps, and aligns the solution with organisational objective [3].

A. Current State:

Right now, The Company "Greenplus" is working in such a way that is making the things harder than they need to be. The Customer information is dispersed over the spreadsheets and the email inboxes, where it means that there is no any single clear view of any customer at any given time. When the complaints come in, there is no proper process for handling or escalating them, so things get delayed and the customers end up frustrated. The Billing is another weak spot where there are arguments which are common because there is little transparency or control over the process, which puts extra pressure on the support team. Though the Marketing is largely guesswork, with no real customer segmentation or any way to measure whether the campaigns are actually working or not. And at the management level, there are no dashboards or the reporting tools in place, so the leaders are essentially making decisions without knowing what is really going on in the business. Taken together, all of these issues point to the same underlying problem, a lack of proper systems, standardised processes, and the data-driven decision making, all of this which is driving up costs, and frustrating customers, and as well as pushing churn higher.

B. Future State:

The aim of the company "Greenplus" is to move towards a way of working that is organised, joined up, and driven by the data. In this future state, everything which is related to the customers like their contracts, the billing history, the support tickets, and the marketing interactions which would all live in one centralised CRM system, giving the team a complete and correct picture of every customer at any time. The Processes that are currently handled manually that would be get automated, and the complaints get dealt with efficiently, the issues get escalated when they need to be, and the overall quality of service becomes far more consistent. The Billing would also be much better controlled, with the monitoring tools as to catch the problems early before they turn into any arguments.

Marketing would shift from guesswork to the targeted, the data-led approach, where the customers are grouped based on their behaviour and the needs, and the campaign results are actually tracked and measured. And the Management would have access to the live dashboards and showing exactly how the business is performing over key areas like churn, service quality, and the customer engagement. so the decisions can be made based on the facts instead of instinct. Although, the Greenplus

would introduce the renewable partnership model, where the data and the analytics are used to identify the customers who have the potential to generate their own renewable energy. And where this would allow the company to grow beyond simply being an electricity supplier and start building a big renewable energy community.

C. Identified Gap:

Comparing where the company "Greenplus" is now ,to where it needs to be which tell us the six clear gaps which needed to be address. so for first, there is a customer data gap as, because there is no any central system which is holding all the customer information, where the records are scattered and inconsistent, as making it very difficult to the manage customer relationships properly. Secondly, there is a complaint management gap, as without any structured process or any escalation path, the complaints are slow to be resolved, poorly tracked, and they are handled differently depending on who picks them up. Third, there is the billing control gap as the problems with bills are only being find, after customers have already complained, rather than being caught early, so this is where it damages the trust and handles up the number of arguments. Fourth, there is a marketing intelligence gap where the campaigns are not properly customized to specific customer groups and also there is no way of measuring whether they are working or not,this leads to poor engagement and wasted effort. Fifth, there is a management reporting gap, because there are no dashboards or any analytical tools in place, where the leaders are making important decisions without dependable. Finally, there is the strategic growth gap for company "Greenplus" currently which has no way of identifying customers who could possibly generate their own renewable energy, that means that the company is leaving a important growth opportunity completely undeveloped.

D. Summary of the Gap Analysis:

The gap analysis makes it clear that the problems "Greenplus" company is facing is not just the few unreachable issues which can be fixed with quick solutions. They run deeper than that as the challenges are rooted in the very way the business is structured, spreading how data is managed, how customer service processes work, and how the analytics are used, as well as how strategic decisions are made. Fixing these things properly requires implementing a centralised CRM system, then automating key processes, and building out the Business Intelligence dashboards that give the management real visibility into the performance. By closing these gaps, the proposed solution aims to make the business run more efficiently, and deliver a better experience for customers, which allow smarter decision making at every level, and lay the groundwork for the renewable partner ecosystem. The intention is not just to put out the fires in the short term, but to build a foundation that supports the long-term growth and sustainability of the business.

8. KPI(Key Performance Indicator) Framework:

The KPI framework is basically the strong stick that will be used to track whether the proposed CRM and Business Intelligence solution is actually making the difference for the Greenplus. As each KPI has been chosen with the specific purpose in mind as together they cover the areas that matter most to the business, including keeping the customer's, resolving the issues faster, and getting billing right, as well as running more effective marketing campaigns, and running the strategic growth. Though Rather than just implementing the new systems and hoping for the best, these KPIs turn the business goals into the real, and the measurable targets so that management can clearly see, what is improving, what still needs to work, and make confident decisions based on the real data rather than the gut feeling.

A. Customer Maintenance KPI (Churn Rate):

The Customer churn rate is one of the most important measures for "Greenplus" as it needs to keep a close eye on right now, as the business is losing customers at a worrying rate. Simply put, it tracks the percentage of customers who leaves the service over a given period of time. It matters so much because it acts as a direct reflection of how satisfied customers are and how stable the business is as when churn is high, it is usually a signal that something is wrong, like whether that is poor service, the billing problems, or a lack of meaningful engagement with the customers.so by tracking the churn through the dashboard over the time, the management can spot patterns,also see whether the customer maintenance efforts are actually working, and when they looked at alongside the other data like complaints and billing disputes, and start to understand the real reasons why the customers are choosing to leave.

B. Service Efficiency KPI:

The average resolution time is simply that how long it will take to sort out the customer complaint or the service which is request from start to finish. As it is an important sign of how well the customer support that the work is running. Right now at "Greenplus" company, the complaints are being handled in an unstructured way with, no clear process in place, which means things are taking longer than they should be and the customers are being left waiting and annoyed. Though, bringing this time down is therefore one of the main things the CRM implementation is aiming to achieve. Where as by tracking this KPI over time, and the management can see whether the new automated workflows and the escalation processes are actually making a difference or not. So when the resolution times start to fall, it is a good sign that the working is becoming more productive and that customers are getting a better experience.

C. Billing Performance KPI:

The billing Performance rate, simply tracks how many bills end up leading to a customer complaint or the arguments. As this is one of the most important examples for the "Greenplus" company to keep an eye on because the billing problems are one of the biggest reasons, the customers are becoming dissatisfied and the ultimately decided to leave. When the arguments rate is high, it is usually a sign that bills are inaccurate, confusing, or lacking the transparency and none of which builds the customer trust. So, by tracking this KPI, the company can see whether the automated alerts and the validation checks which is being introduced through the CRM system are actually making a difference to the billing accuracy over the time. It also acts as an early warning system, that helping the team spot patterns or the systemic problems in the billing process before they spiral into something bigger and harder to fix.

D. Marketing Effectiveness KPI (Conversion Rate):

Conversion rate is simply means that how well the marketing campaign is actually working, in other words, it is like how many customers are taking the action for the campaign which was designed to encourage, whether that is signing up, and staying with the company, or getting involved in a particular programme. Though right now, the Greenplus's marketing efforts are not backed by data and there is no proper way of tracking whether the campaigns are having any real impact. So, this KPI changes that by giving the company a clear way to see what is working and what is not. Once, the campaign data is pulled into the CRM and reflected in the dashboards, the management can start to understand that, how many customers are responding and use that information to make the future campaigns more targeted and relevant. The higher the conversion rate, the better the communication and the stronger the relationship with customers becomes.

E. Strategic Growth KPI (Renewable Partner Leads):

One of the new strategic measures which is being introduced through this project is the tracking of the renewable partner leads which is important, the number of customers who are identified as the good candidates to become the active participants in the "Greenplus's renewable energy" partnership model. This KPI is important because it reflects how well the company is progressing, beyond its role as the straightforward electricity supplier and the moving towards building a wider renewable energy community. The customers are identified through a data driven scoring process which looks at factors like where they live, how much energy they use, and how engaged they are with the company. So, by keeping an close eye on this number, the management can monitor how the renewable partner pipeline is growing over the time and get a sense of whether the campaigns which is designed to encourage the customer participation in renewable energy generation is actually having an impact or not.

F. Importance of the KPI Framework:

The KPI framework is playing an important role for the changing the Greenplus Energy Ireland into a data driven organization. So, instead of depending on the assumptions, the management can use these measures to monitor the business performance in real time and to identify the problems area quickly, where as to evaluate the impact of the implemented solutions and to support the strategic decision-making. Though by linking each KPI to a some specific business objective, the framework make sures that the CRM and BI solution to deliver the measurable value to the organization.

9. Tool Selection and the Justification:

A. Salesforce (CRM):

Salesforce has been chosen as the CRM platform because it gives the company "Greenplus" the ability to properly organise and to manage the customer data in one place. Also it supports the custom data structures, the automated workflows, and the controlled access, that means the right people can see the right information and the routine processes which can run without having everything to be done manually.

B. Power BI for Dashboard:

The reason to take the Power BI as a tool because it is well suited to turning the data into clear, visual insights that management can actually use. Also, it offers the strong data modelling, the ability to track the key performance indicators, as well as the interactive dashboards that make it easy to monitor how the business is performing and make the informed decisions quickly.

10. As-Is and To-Be Process Analysis:

A. Current Process(A-Is):

Right now, when the customer raises an issue, it gets records manually and it gets recorded at all. From there, the response is slow, the follow-up is inconsistent, and many of the complaints end up going unresolved. Although the end result is a frustrated customer who is far more likely to leave.

B. Proposed Process (To-Be):

Here, in the new process, when the customer raises an issue, the ticket is automatically created in the CRM system. so, that ticket is immediately prioritized based on its urgency, and if it needs to be escalated, that happens automatically too. The issue gets resolved in a structured and timely way, the customer is asked for feedback, and the whole interaction is tracked and visible on the management dashboard.

C. Renewable Partner Process (To-Be):

On the strategic side, the customers would be run through an analytical scoring model that assesses their potential to generate the renewable energy. Those who score highly that would be signaled as the partner leads and then will get entered into a targeted campaign which encouraging them to consider for the solar installation. Once they install and begin to generating energy, their performance would be tracked over time, creating a growing network of renewable energy contributors within the Greenplus customer base.

11. Methodological Approach:

This project follows the CRISP-DM methodology, which is basically a step-by-step framework that actually guides the work all the way from the very first, understanding the business problem through, to actually delivering and deploying the solution. Alongside this, a range of well-established business analysis techniques are used which is including the stakeholder analysis, SWOT analysis, gap analysis, and the process mapping, so to make sure the requirements are properly defined and grounded in the reality. Using this structured approach make sure that everything developed throughout the project, whether it is an operational improvement or an analytical information, it is directly matched to what the business actually needs to achieve.

12. Database Design and Data Model:

A. Purpose of the Data Model:

The data model is basically the blueprint which determines how all of the business's information is organised, stored, and connected within the system. Though getting this right is important because it makes sure that every key process in the business whether it's from managing the customer accounts and handling the billing, to resolving complaints, running the marketing campaigns, and identifying the potential renewable partners, it is properly supported and runs smoothly.

For company "Greenplus" specifically, the data model needs to do two things at once. It needs to support the day-to-day operational side of the business, such as billing, complaint handling, and customer relationship management, while also supporting the analytical side, such as understanding, why customers are leaving, measuring service performance, and spotting the customers who could become the renewable energy partners. To achieve this, the design takes the structured approach where each important part of the business is represented as its own table, and the clear relationships are set up between those tables to keep everything consistent and make the meaningful analysis possible.

B. Core Entities in the System:

The system which is being proposed is built around the main things that actually matter to the day-to-day running of the company "Greenplus". As there are eight core building blocks such as Customers, Contracts, Bills, Support Tickets, Campaigns, Tariff Plans, Renewable Partners, and the Energy Production. Each of these return a real part of how the business operates, and together they give a complete picture of everything from how the customers are being billed and supported, to how marketing campaigns are performing and how the renewable energy partnerships are developing. So, having all of these connected in one system also makes it possible to carry out more of the strategic analysis, such as tracking that why customers are leaving, measuring how well complaints are being handled, and identifying the opportunities to grow the renewable partner side of the business.

The system is designed around the following key entities such as :

Customers - Stores the core customer profile and segmentation data

Contracts - Stores tariff agreement and contract duration details

Bills - Stores billing amount, consumption, payment, and dispute status

Support Tickets - Stores customer issues, status, priority

Campaigns - Stores marketing campaign and engagement information

Tariff Plans - Stores pricing structure and contract plan details

Renewable Partners - Stores customer partnership status and solar participation.

Energy Production - Stores mostly generation and supply data for the partners.

These entities describes the core operations and the strategic components of the business.

C. Entity Relationship Design (ERD Explanation):

The Entity Relationship Diagram was created to show how the data flows through the business and connects over the different stages of the customer's journey with the company "Greenplus". At the centre of the whole model is the Customer, which makes sense, because almost everything the business does, it either starts with or connects back to an individual customer. Though from there, the model branches out to reflect the real life like, a customer can have one or more contracts, receive multiple bills, raise support tickets, and engage with the marketing campaigns. As, there is also an optional path where the customer can become a renewable energy partner, in which case their energy generation activity is recorded in a linked table. The whole design shows the way Company "Greenplus" actually operates, recognising that customer data, the billing, the support, and the renewable participation that are all connected to each other rather than existing in separate silos.

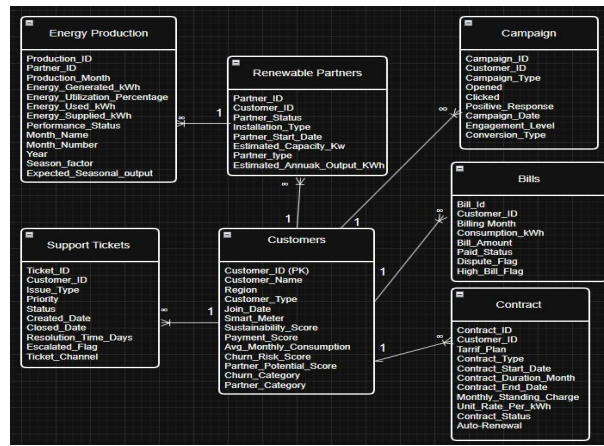


Fig 1: Entity Relationship Design

Figure 1 shows that the logical data model for the proposed system. In the center, of the diagram is the customer, which though makes sense given that everything the company "Greenplus" do eventually in the end, it revolves around its customers. From there, the model connects out to all of the key operational and the strategic areas of the business. This structure, it has been designed to support both the CRM side of things and handling day-to-day customer transactions and the interactions whereas, the Power BI reporting side, which gives management the visibility that they need to track the performance and make the informed decisions.

C. Relationships Between the Entities:

The way the data is structured shows, how the business actually works in the real life. A single customer can receive many bills over the time, so the system is designed to link multiple bills back to the one customer. In the same way, a customer might raise several support tickets at different points, and whereas, all of those are connected back to their customer record. Though a customer can also hold more than one contract, and each contract is tied to a specific tariff plan. The Renewable partnership is treated as an optional add on to the customer profile as not every customer will become a renewable partner, but for those who do, the system can track multiple monthly energy production records which is linked to that partner. This structure keeps the data clean and consistent while making it possible to analyse how the different parts of the business connect and influence each other.

E. Data Model Type:

The system which is being proposed uses a two-part approach to the handling data. As, inside the CRM, the data is stored in a clean, structured way where everything is properly linked and organised, so the records like customer profiles, contracts, bills, and the support tickets are all consistent and accurate. But when that data needs to be used for the reporting and the dashboards in Power BI, it gets transformed into a format that is much easier and faster to work with for analysis. Though, this reporting structure is built around the clear categories such as customer details, the dates, the regions, the tariff plans, and the partner status, and making it straightforward for the dashboards to pull up the insights quickly and present them in a way that is easy for the management to understand and act on.

14. Data Dictionary:

A. Purpose of the Data Dictionary:

A data dictionary was put together to clearly define that, what each piece of the data in the proposed system actually means, as where it comes from, and what it will be used for. This make sure about the consistency in implementation and supports both for the CRM configuration and dashboard modelling. As well as the Data Dictionary also provides a clear reference for key identifiers, calculated fields, and relationships over the tables.

B. Logical Data Model or Table Schema:

1) Table Name: Customers:

Field Name	Description	Data Type
Customer ID	Unique identifier for each customer	Int (Primary Key)
Customer Name	Full name of Customer	Text
Region	Customer location	Text
Customer Type	Residential or Small Business	Text
Join Date	Customer Joining Date	Date
Smart Meter	Do customer have smart meter or not	Boolean

Sustainability_Score	Shows customer alignment with sustainability values	Int
Payment_Score	Showing payment reliability	Int
Avg_Monthly_Consumption	Average electricity consumption per month in kWh	Decimal/Number
Churn_Risk_Score	Calculated score showing risk of customer's leaving	Int
Partner_Potential_Score	Calculated score showing renewable partner suitability	Int
Churn_Category	Categorize churn risk level Eg: Low, Medium, High	Text
Partner_Category	Categorized renewable partner potential Eg: Low, Medium, High	Text

2) Table Name: Contract Table:

Field Name	Description	Data Type
Contract_ID	Unique identifier for each contract	Int
Customer_ID	Identifier linking the contract to a customer	Foreign Key
Tariff_Plan		
Contract_Type		Date
Contract_Start_Date	Date on which the contract begins	Date
Contract_Duration_Month	Contract duration month	Number
Contract_End_Date	Date on which the contract End	Date
Monthly_Standing_Charge		
Unit_Rate_Per_kWh		
Contract_Status		
Auto-Renewal		

3) Table Name: Bills Table:

Field Name	Description	Data Type
Bill_Id	Unique identifier for each bill	Int
Customer_ID	Identifier linking the bill to customer	Int
Billing_Month	Month to which the bill applies	Date
Consumption_kWh	Electricity consumption during the billing period	Decimal/Number
Bill_Amount	Amount charged for the billing period	Decimal
Paid_Status	Tells Whether the bill was paid	Text
Dispute_Flag	Shows whether the customer disputed the bill	Boolean
High_Bill_Flag	Shows whether the bill is classified as unusually high	Boolean

4) Table Name: Support Tickets Table:

Field Name	Description	Data Type
Ticket_ID	Unique identifier for each support ticket	Int
Customer_ID	Identifier linking the ticket to a customer	Int
Issue_Type	Type of issue raised Eg: Billing, Complaint, Service, Meter Issue	Text
Priority	Ticket Priority level Eg: Low, Medium, High	Text
Status	Current status of ticket Eg: Open, Closed, Escalated	Text
Created_Date	Date when ticket was created	Date
Closed_Date	Date when ticket was closed	Date
Resolution_Time_Days	Number of Days taken to resolve the ticket	Int

Escalated_Flag	Indicates whether the case was escalated	Boolean
Ticket_Channel	Channel through which the issue was raised. Eg: Email,Phone,Web	Text

5) Campaign Table:

Field Name	Description	Data Type
Campaign_ID	Unique identifier for each campaign interaction record	Int
Customer_ID	Identifier linking the campaign record to a customer	Int
Campaign_Type	Type of Campaign Eg:Email,Retention,Renewable Outreach, Smart Meter Awareness	Text
Opened	Tells whether the customer opened the campaign	Boolean
Clicked	Shows whether the customer clicked campaign content	Boolean
Positive_Response	Tells Whether the customer responded positively	Boolean
Campaign_Date	Date the campaign was sent or recorded	Date
Engagement_Level	Campaign engagement level Eg: Low, Medium,High	Text
Conversion_Type	Tells whether the campaign led to a desired action	Boolean

6) Renewable Partners Table:

Field Name	Description	Data Type
Partner_ID	Unique identifier for each renewable partner record	Int
Customer_ID	Identifier linking the renewable partner to a customer	Int
Partner_Status	Current partner status Eg: Potential, Interested, Converted, Active	Text
Installation_Type	Type of installation Eg: Rooftop or Land-based	Text
Partner_Start_Date	Date the renewable partnership begins	Date
Estimated_Capacity_Kw	Estimated electricity generation capacity in kilowatts	Decimal/Number
Partner_type	Type of Partner Eg: Residential Solar, Small Business Solar	Text
Estimated_Annual_Output_KWh	Estimated Annual electricity generation capacity in kilowatts	Number

7) Energy Production Table:

Field Name	Description	Data Type
Production_ID	Unique identifier for each energy production record	Int
Partner_ID	Identifier linking the record to a renewable partner	Int
Production_Month	Month in which the energy was produced	Date
Energy_Generated_kWh	Total electricity generated in kilowatt hours	Decimal
Energy_Utilization_Percentage	Electricity utilization percentage	Decimal
Energy_Used_kWh	Energy used in kWh	Number
Energy_Supplied_kWh	Electricity supplied back to the grid in kilowatt hours	Number

Performance_Status	Shows the Energy performace Eg: Low, Medium, High	Text
Month_Name	Month Name	Text
Month_Number	Month Number	Number
Year	Month Year	Number
Season_factor	Electricity seasonal factor	Decimal
Expected_Seasonal_output	Expected seasonal output	Decimal

15. Mock Data Design and Pre-Processing:

A. Need for Mock Data:

As the company "Greenplus" Energy Ireland is the fictional company, so there was no real operational data available to work with it. To get around this, the mock datasets were created to recreate what realistic business data would actually look like for the company of this type. Though, the data wasn't just randomly generated as it was carefully designed to think about that the believable patterns over areas like customer behaviour, billing activity, service issues, marketing campaign engagement, and the renewable partner activity. The purpose of these datasets was to support the building and testing of the CRM system, the Business Intelligence dashboards, and the various business workflows that form part of the proposed solution.

B. Datasets:

Datasets Table:

Dataset	Purpose	Row Count
Customers	Customer records	1000
Contracts	Customer agreements	1000
Bills	Monthly billing records	3000
Support Tickets	Complaint and service records	1000
Campaigns	Marketing engagement idea	1000
Renewable Partners	Active Partner record	350
Energy Production	Monthly generation records	

Total Seven main datasets were created to show the operational and the analytical needs of the business. The selected number of rows volumes were large enough to support the meaningful dashboard analysis.

C. Mock Data Logic:

To make the mock data as real and useful as possible, the logical connections were built in between the different data points to show how the things would actually work in the real business. For example, the customers who use more electricity naturally have higher bills. Where billing exception exist, there is a higher chance that a argument has been raised. Customers who have contacted the support multiple times are signalled as being at the greater risk of leaving. And where as, the customers who show a strong interest in sustainability, where they pay their bills reliably, and are based in the suitable locations for them are given higher scores as the potential renewable energy partners. These built in the relationships mean that the data behaves in a way that mirrors real business conditions, making it far more meaningful and valuable for the analysis rather than just being a random set of numbers.

D. Data Generation Method:

The sample data was used in this project, as it was created using an online data generation tool which is called Mockaroo, and then itis tidied up manually to make sure that everything is linked together properly over the different datasets. Because the tool has the limit on the rows like how many rows could be generated at once for example maximum 1000, so, the data was exported in several batches and then it was combined into the larger, unified files. This approach made it possible to work with a real volume of the data while keeping the structure consistent throughout.

E. Data Pre-processing steps:

Before the data could be use for the Salesforce and the Power BI, there was some important steps for work that was carried out to make sure that everything was clean, consistent, and ready to use. where as, the some steps involved filling in or addressing the missing information, and also making sure that the dates and other values were formatted the same way throughout, and checking that records were properly linked to each other properly. On top of that, there were some additional calculated fields that were created, such as scores to indicate that how likely a customer is to leave, as well as how suitable they might be as a renewable energy partner, and how long the complaints took to be resolved. All these steps was necessary

to make sure that the data would work properly for both the day-to-day customer management side of the things and the reporting and the analytics side.

16. Exploratory Data Analysis(EDA):

A. Purpose of the EDA:

The Exploratory Data Analysis was carried out to understand the structure and behaviour of the mock data that would be used in the system, before building anything in Salesforce or Power BI. To look for the data first and understand that what it was showing and where as spotting the trends, the patterns, and the connections that are relevant to what the business is trying to achieve. The main areas of focus were the customer churn, billing behaviour, and the service-related indicators, since these are the things that most directly affect whether the customers stays or leaves, and also how smoothly the business operates day to day.

B. Churn Category Distribution:

To understand the customer retention risk within the dataset, the churn category distribution analysis was carried out. Where the customers were grouped into Low, Medium, and High Churn categories based on there calculated churn risk scores.



Fig. 8 Distribution of Customers by Churn Category

The figure 8 shows the Distribution of customers over the three churn categories. The Analysis shows that the customers falls into the medium churn category, which tells that a large segment of the customer base is not yet lost but it remains helpless. This tells that the company "Greenplus" needs to take a more proactive approach to identify that which customers are at risk of leaving and reaching out to them before they do. So from the Business point of view this finding shows exactly why a proper CRM system is needed, one that can flag these at-risk customers early and allow the team to take the right steps to keep them.

D. Relationship Between Consumption and Billing

To check whether the billing dataset is consistent and make sense, the analysis was carried out as the relationship between how much electricity customers use and how much they are charged. As it is expected that customers with higher electricity consumption should pay more bills. A Scatter plot was used to visually see this relationship.

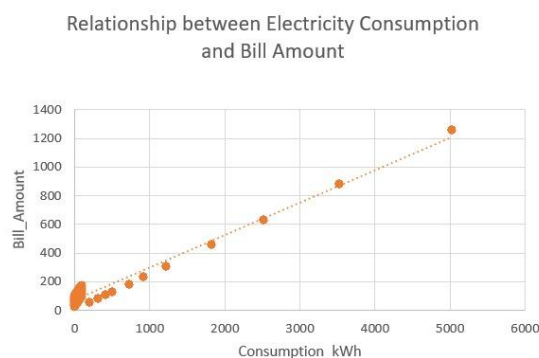


Fig 9. Relationship Between Electricity Consumption and Bill Amount

The Figure 9 shows that, there is a clear and consistent relationship between how much electricity the customer uses and how much they are billed, where as the more they consume, the higher their bill is, which is exactly what you would expect in the real world. As Most of the data sits at the lower end of the consumption scale, showing that the typical household customer who uses a moderate amount of electricity, while a smaller number of higher usage cases likely shows the heavier users or

small business customers. This shows that the dataset feel realistic, since energy usage in the real world is never perfectly uniform. The near-straight-line pattern also confirms that the billing figures have been logically calculated from the consumption levels, making this data reliable and useful for the analysing billing performance, spotting the high-bill customers, and identifying the dispute patterns.

D. Key Findings from the EDA:

The exploratory data analysis confirmed that the mock data is real and fit for the purpose. It also showed that the meaningful number of customers are at risk of leaving, which highlights the importance of having the proper retention strategies in place. Consumption and billing patterns also lined up in the logical way, which is giving the confidence that the data accurately shows the real-world behaviour. Though, these findings confirm that the datasets are solid enough to be used as the foundation for building out the CRM configuration and Power BI dashboards.

PART 2 : IMPLEMENTATION REPORT

1. Development Process:

The project started by identifying key business challenges in the company, specifically regarding customer management, complaint tracking and not having a clear view of renewable energy opportunities. The reason behind developing a system was to create a centralised location and to collect all of the company's customer data to be used for the decision and the supporting of it with the structured information. The process was methodical, and the first step was to generate the sample data that correctly represented the real individuals. The development of the CRM and Business Intelligence solution follows a structured approach which is inspired by the CRISP-DM methodology which highlights the business understanding, data preparation, modelling and evaluation [3]. Whereas, the sample data were formatted and cleaned up so that they could be compared by several different criteria, for instance, customer types, billing information, and location (geographic). Following the clarification of the structure of the data collected, the CRM system was then being configured using the Salesforce application. Initially, the design was to create multiple structures for each department with the use of custom structures. However, when the implementation of the system was completed, it became apparent that it would be much easier and practical to use existing structures within the system such as customer profiles, support cases and the opportunity tracking. Therefore, the final configuration of the system was improved and would lead to a less complicated system design. While the development of the system continued, an additional issue became evident that customer complaints were being tracked but not being recorded for an urgency basis. As a result of that, an automated escalation process had to be implemented towards the end of the development cycle. This was not included in the original design of the system.

A. Implementation Architecture:

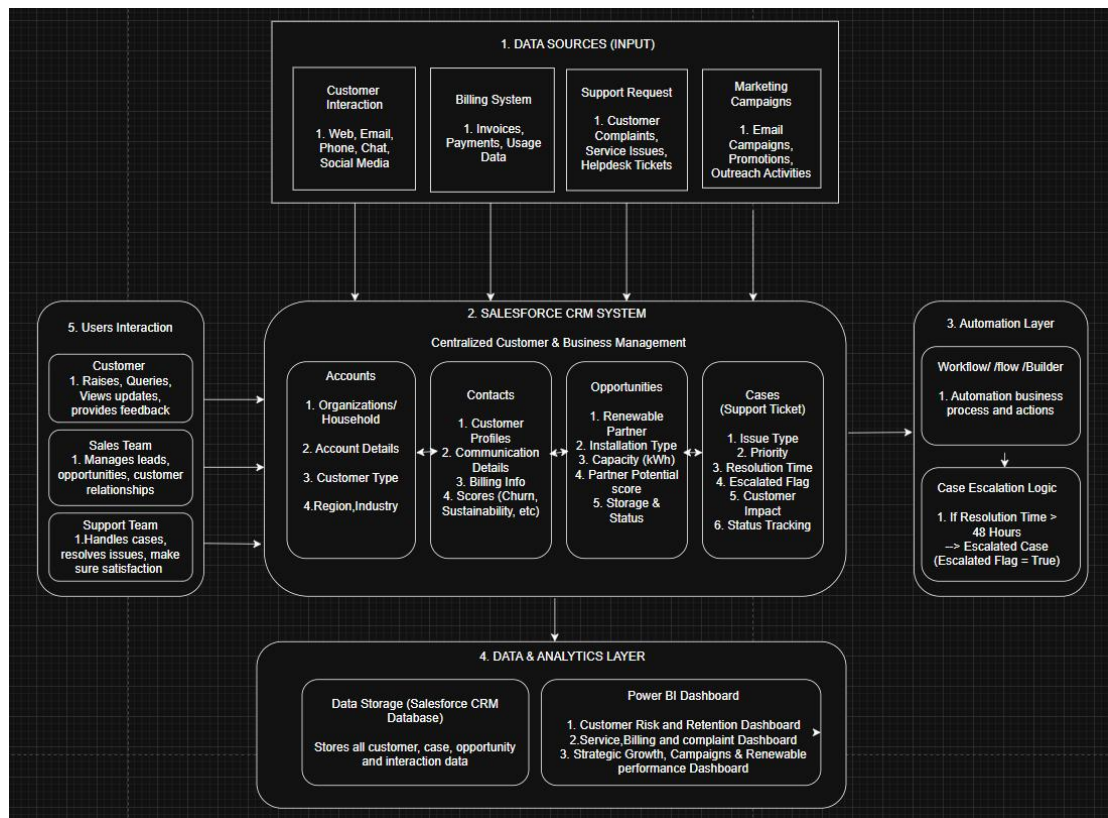


Fig 10: Implementation Architecture

2. CRM System Implementation:

As the system's architecture diagram shows, the CRM software is designed to combine, integrate, and manage all customer information for the operational and the business processes efficiently. The system has been divided into the several levels of the layers to show how different types of information generated from the inputs move through each of the supporting elements of the system to produce the outputs to support the decision decisions made by the management. On the top level, the company receives data from the different types of sources, including the customer interactions like phone calls, chat, or the email, billing systems, support requests, and marketing activities. This is the primary data entry point the organisation uses for all major customer and operational data capture. Though, all major customer and the operational data will be captured in one system rather than being separated, as was often done in the past. The Salesforce CRM is the heart of the system architecture and is the system that provides the single location for the organisation to manage all its business data. The

Salesforce system uses the several different types of the components to organise and struct the data like accounts, contacts, opportunities, and the cases. The accounts and the contacts components store the customer data, the opportunities component tracks the possible renewable energy projects for each customer, and the cases component helps manage any issues or service request from the customers. Although, this helps provide an organisation with a unified organisation of customer information, which gives improve visibility over the customer data and provides the organisation with the comprehensive view of its customers. In addition to the above components, the CRM system’s architecture also has an automation layer supporting the processes used throughout the organisation to increase productivity.

B. Overview of the CRM Design:

The CRM (Customer Relationship Management) it was created to help a company to keep the entire history of customer interactions in one location. It manages all information about each customer, including managing customer complaints, tracking possible renewable energy projects, and supporting communications through marketing campaigns. The system is created around the customer where each customer has an account associated with them. Customer complaints are recorded as “Cases”, while the business opportunities to customers are kept track of in their own section. This allows for a more organized and connected system rather than having data placed in a more than different locations.

C. Customer and Account Management:

The first step of the system is to create the customer profiles and linking them with their respective accounts. Each account is classified based upon whether it is a house hold or the business account. As each of the contacts linked to these accounts represents an individual customer. As has been told previously, there are many customers represented by these accounts. Examples of some of the customers are, Emma Walsh, Sarah O’Brien, Liam Doyle, Conor Kelly, and John Murphy. Each of these accounts also contains multiple attributes associated with the account like the attributes which can be further classified based upon regional location, account type, billing information, and credit risk ratings. The structure of the system allows for all customer-related information to be maintained and retrieved through the system based upon the customer ID as the main index key for accessing customer information records.

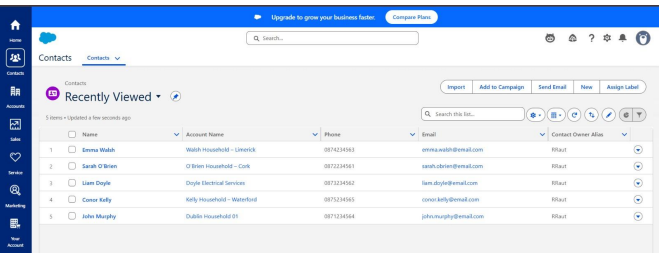


Fig 1 : Customer Contact Details

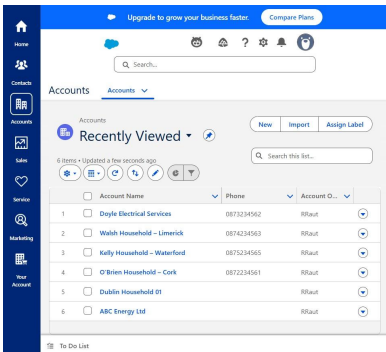


Fig 2: Customer Account Details

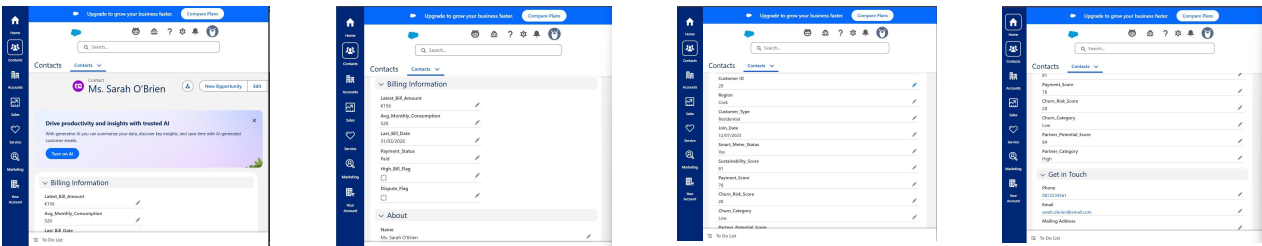


Fig 3: Customer profile

The CRM system was developed with a primary goal of creating the structured and meaningful representation of the customer data. This was achieved by using the Accounts to show Households or Small Businesses, and using Contacts to represent an individual customer and linking them to their corresponding Account. The information from the Accounts List indicates that there are multiple groups of customers that have been established as a result of the system being created to support many different customers. This is critical to make sure that we understand the different needs of our customers, as well as the behaviours of each customer. Each customer is linked to an Account in the Contacts List and it has the corresponding contact details such as phone and email which is associated with them. This allows us to have all of the information pertaining to the customers in one location and not have it fragmented within numerous other systems. The detailed Contact Record shows us to obtain a broader depth of the knowledge regarding each customer. The information included in this area includes the billing information such as the most recent bill and whether or not the bill has been paid, as well as some analytic information such as the customer's churn risk score and sustainability score. This shows us that the system not only maintains customer data but also assists us with the business analysis. Based on the above, the solution for the lack of visibility with regard to customers is addressed by having a single structured view of each customer by way of the system.

D. Complaint and Case Management:

In the System, each case contains different types of information including type of the issue, priority level, amount of time required for resolution, and how the issue affects the customer. Another important function of the system is tracking the length of time it takes to resolve a customer complaint, which helps the organisation to detect any delays and subsequently resolve them in a timely manner.

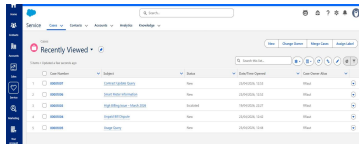


Fig 4: Showing Multiple Cases

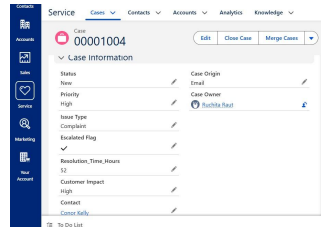


Fig 5: Case having Fields

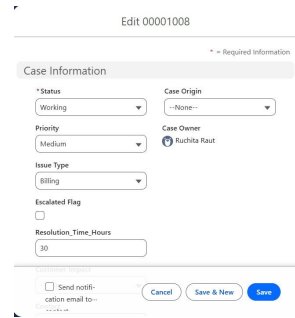


Fig 6: Case Record
(Escalated=False,30 hr)

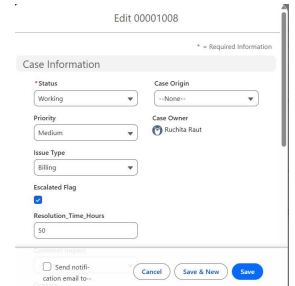


Fig 7: Escalation After
(Escalation=True,52 hr)

To provide an organised method for resolving the customer complaints, The Case Management feature was created. When the customers have an complaint (for example like, a service problem, a billing problem, etc.) each complaint record is known as a case. The case list contains of the multiple complaint records which indicate that the current status and the subject of each complaint. This gives an overview of all customer-related issues to make management and the monitoring of these issues easier. An example of an escalated case is when the expected amount of time to resolve an issue has exceed what was expected, so the system automatically deems this escalated that means it needs an immediate attention. In opposite, the non-escalated case is a normal issue that has not yet exceeded the expected amount of time to resolve. The fields configuration screen displays the field(s) that were added to the track cases, which include item type that is the reason for the case, time to resolve the case, an escalated flag, and the customer impact of each case. This will allow for consistent recording of all cases for improved the tracking and analysis. The results of implementing a case management solution greatly improve the customer complaint resolution process by the improving how organised, transparent, and measurable the complaint resolution process is.

E. Opportunity Management (Renewable Energy Focus):

The system tracks the possible new sources of the renewable energy. The customers included in this portion of the system could be installed with solar systems or receive other forms of energy upgrades. Each of these opportunities captures the information related to the potential capacity of the solar project, each type of installation, and the expected project value. This

information allows the company to determine which customers are likely to assist with the company’s long-term growth objectives.

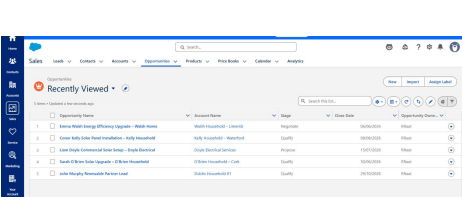


Fig 8: Fileds of Opportunity

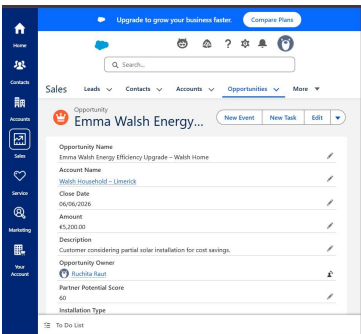


Fig 9: Customer detailed Opportunity

As Businesses can use the opportunities in the Salesforce to track the potential Renewable Energy Partnerships. This make sure that the company can find customers who may be interested in solar installations or the energy upgrades. The opportunities list shows the multiple different potential projects that are accessible to the customer account. This allows for a pipeline view of the potential future business opportunities available. Within each individual opportunity record, there are detailed fields which include the estimated project value, installation type, and the partner potential score. An example of this would be the Large Commercial Solar option for Liam Doyle, which has the high estimated project value rating that indicates that this is a important business opportunity. The configuration of the opportunity fields includes the custom field additions specific to tracking the partner status and the estimating capacity. As such, the company should track opportunities to help to measure against the potential return on investment as well as to identify priority opportunities. This aspect of the system gives the business the ability to support its growth by tracking and identifying the new sources of the revenue

E. System Configuration and Data Structure:

To add an path of the business analytical support to the configuration, additional fields were added to the system. Billing, customer risk, sustainable score, and partner potential were all examples of this. The relationships between the different parts of the system were defined. For example, links were created between customers and accounts as well as between the cases and customers and between opportunities and the accounts. As a result, the structure of each of these business functions will be formatted correctly, which is allowing them to be used for operations as well as supported in the future by the analysis of that same business function's data.

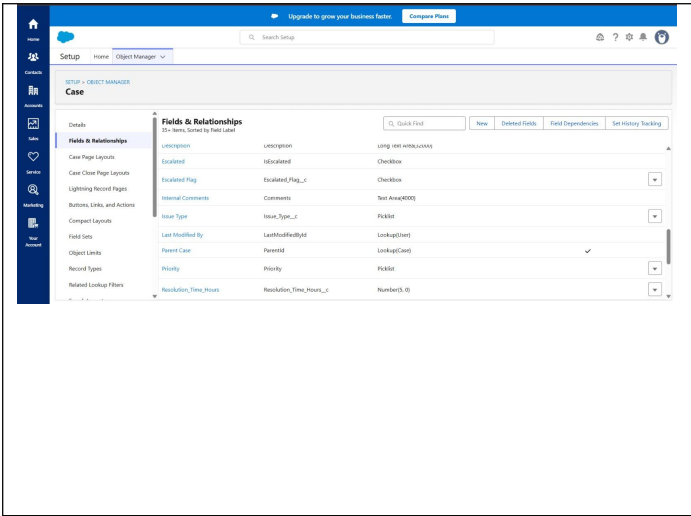


Fig 10: Case fields & relationships screen

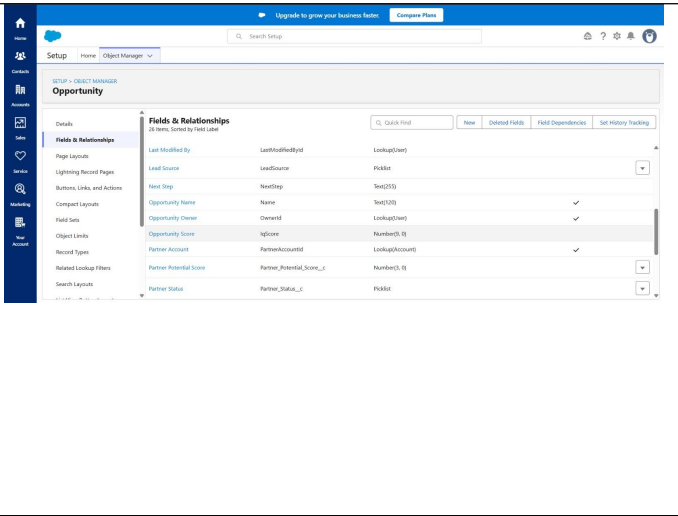


Fig 11: Opportunity fields & relationships screen

The System includes the defined data structure with various custom fields that will contain relevant business info through its use of databases. These custom fields allow relevant data to be stored in the same manner, as well as analysed easily later. For example, the 'Case' object contains the following custom fields like, Issue Type which means (the type of issue being resolved), Resolution Time which means (the time it took to resolve the issue) and Escalation Status which means (whether the case was escalated). As well as the 'Opportunity' object on the other hand, contains the following custom fields like the Partner Potential which means (the potential value of the partner) and Installation Details that means (the details related to the installation). The relationships established between "Objects" (e.g., a contact linked to an account, case linked to a contact) allow us to connect all the data together, thereby allowing us to accurately track the customer activity and understand the complete customer journey. Using this method of organizing data provides the greater data accuracy and allows for a better ability for the business decisions based on the known information.

G. Automation and Workflow:

One of the most important key upgrade to the system was when an escalation rule for complaints was implemented. If a complaint is unresolved for over 48 hours, it will automatically be flagged as escalated. As a result, the tracking complaints will no longer need to be done manually. This will help guarantee that the significant complaint issues do not go unnoticed.

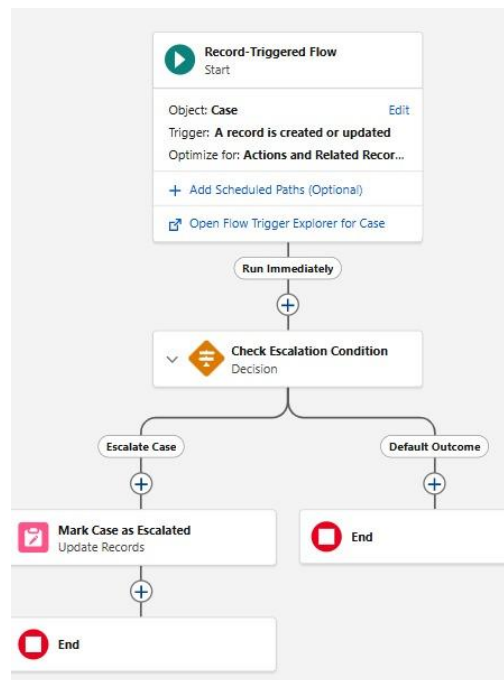


Fig 12: Full flow diagram

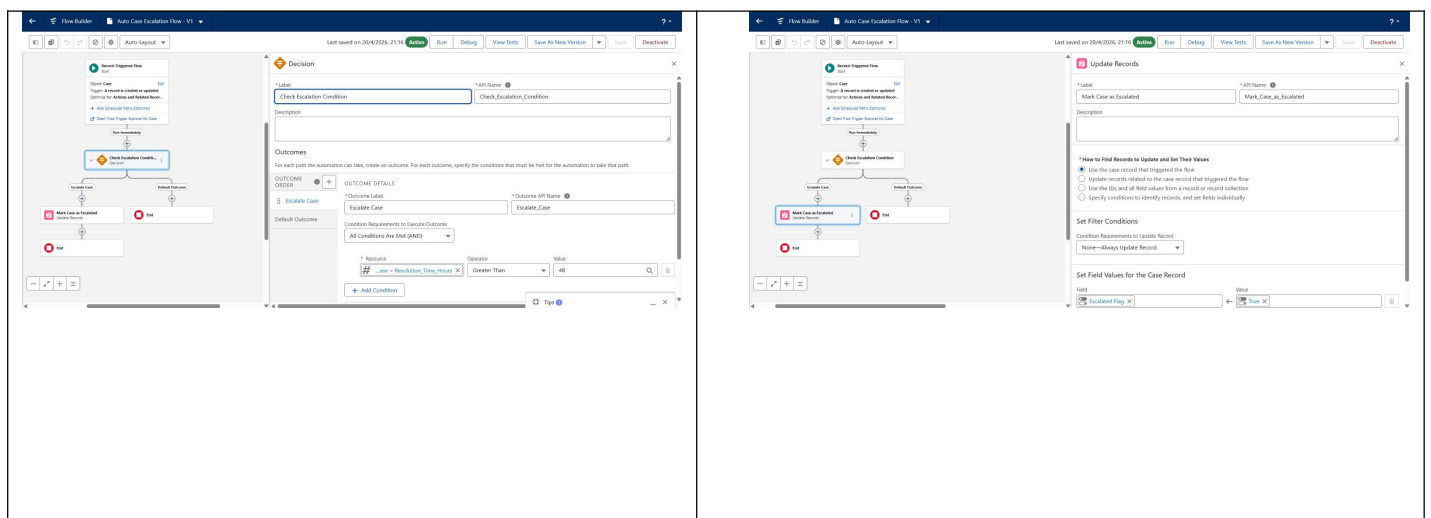


Fig 13: Decision Condition (Resolution Time > 48)

Fig 13: Update step (Escalated Flag = TRUE)

The Automated workflows was implemented to enhance the customer complaint handling. This process is designed to detect the resolution times for each of the complaint case and to automatically escalate them if more than 48 hours has exceeded from creation or modification. The process flow diagrams shows the overall process, beginning with a trigger that occurs when a new either new or existing case is created or modified. The Movement between the standard stages of this process also requires the decision point where there is the comparison between the calculated case resolution time and the defined threshold. If the calculated case resolution time is greater than the defined threshold, the automated workflow will update the case and assign an 'escalated' status. This will eliminate the need for manual case tracking and help to make sure that all significant issues are managed. The last image shows what an example of this type of automation looks like, where a case has been automatically identified as having been escalated. Though, this function improves the overall operational effectiveness, resulting in the faster response time to resolve the customer related problems.

Once again, if any customer complaint goes unresolved after 48 hours, the automation will mark it as escalated to allow timely resolution and better communication with customers. For example, a billing issue remained unresolved for 52 hours and was flagged as escalated in the system, making sure that no manual tracking is needed for critical issues while making the process faster to respond to customers. Therefore, the automation allows for the quicker response time and additionally creates a better experience for the customer overall.

3. Power BI Dashboard Implementation

A. Data Source and Integration Approach:

The The Business Intelligence tool such as power BI allows organizations to transform raw data into interactive dashboards, supporting the real-time monitoring and the analysis of key performance indicators [4]. These dashboards allows users to look for the more trends, identify performance issues, and make data-driven decisions efficiently [4]. Power BI dashboards were created using mocked data from mock datasets that were developed for the business scenario. The CRM has a very small number of records those are manually created to demonstrate how the information is entered into the system, and although the CRM will not have a large number of records, the mock dataset can provide the meaningful analysis on the large scale so that realistic patterns like customer churn, billing problems, service performance, etc. can be analysed in a way that could not occur with a small number of records. The structure of the mocked data continues to match the structure of the CRM data model, which make sure that there will be an alignment between the operational system Salesforce CRM and the analytical system Power BI.

A. Dashboard 1 - Customer Risk and Retention:



Fig 14: Customer Risk and Retention Dashboard

According to the dashboard which is created with the Microsoft Power Bi shows that there are a total of 982 customers over all of the brands of business for this company. There have been identified as having a high risk for churning 95, or roughly 9.7% of the customer base. A very concerning detail about this segment is that this is the customer segment that requires urgent action. The distribution of customers over the various segments shows that most of the customer base is classified as low risk or the medium risk, however, there are also a considerable number of customers which is classified as high risk for churning. While the statistics for the overall customer base could show the stable situation, there is a large concentration of the risk that could negatively impact the company's revenues. An additional analysis shows that high-risk churn customers are primarily located in the Kilkenny, Mayo, Galway and Cork. This could show that the risk is based upon certain locations and not randomly scattered, and therefore, may be attributed either to quality of service, pricing, or operational inefficiencies within these geographic areas. In addition to this geographic dependence, another finding is that the relationship between the churn risk and potential for partnership. The potential to generate the important revenue from customers at a high risk of churning and also being the highest potential partners provides this company with tremendous opportunity. To lose these customers would not only result in loss of revenue, it would finish the future opportunities for the growth as well.

B. Service, Billing & Complaints:

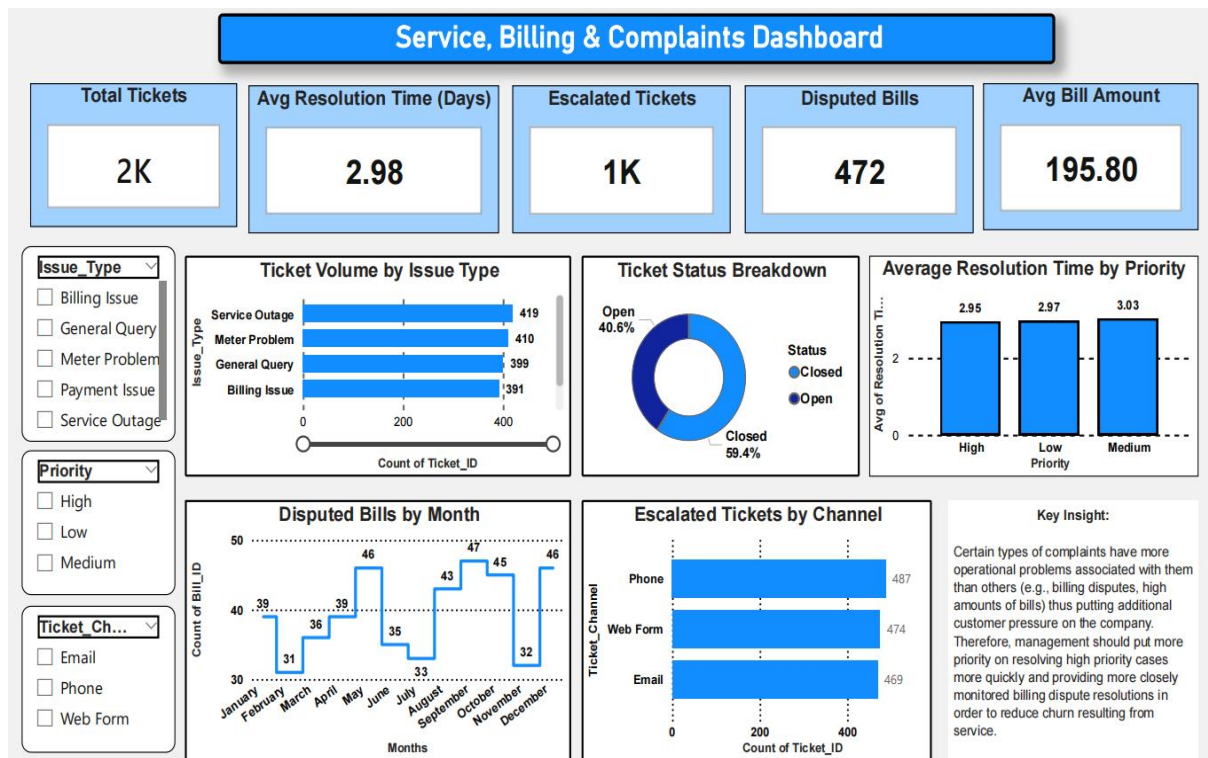


Fig 15: Service, Billing & Complaints Dashboard

The performance of operations around the provision of the services to customers and billing is found in the second dashboard to support the operational analysis. The organization currently has roughly two thousand service tickets to process, however, a large number of cases are described as being escalated. An average resolution time of approximately 2.98 days may seem reasonable, however, the important number of the escalating tickets presents the evidence that processes in staffing complex issues are inefficient. This is the direct relationship to the CRM automation system where the automatic escalation of cases exceed a predetermined threshold. Billing types of issues are one of the most frequently reported types of the service issues as well service outages and the meter-type issues. Though, there are 472 disputed bills, indicating that billing accuracy and transparency are key issues affecting the customer satisfaction. An analysis of the source of the types of complaints have shown there's a fairly even distribution of both phone and email and on-line forms used to report the issues. The type of complaint is not based on the type of communication but on the service or billing processes. From the business perspective, this dashboard serves to show the operational gaps in the organization today. If billing disputes and other high-priority case aren't being resolved efficiently, then those are going to add to customer churn. Therefore, priority should be placed on improving the response time and resolving the billing complaints.

C. Strategic Growth & Renewable Performance:

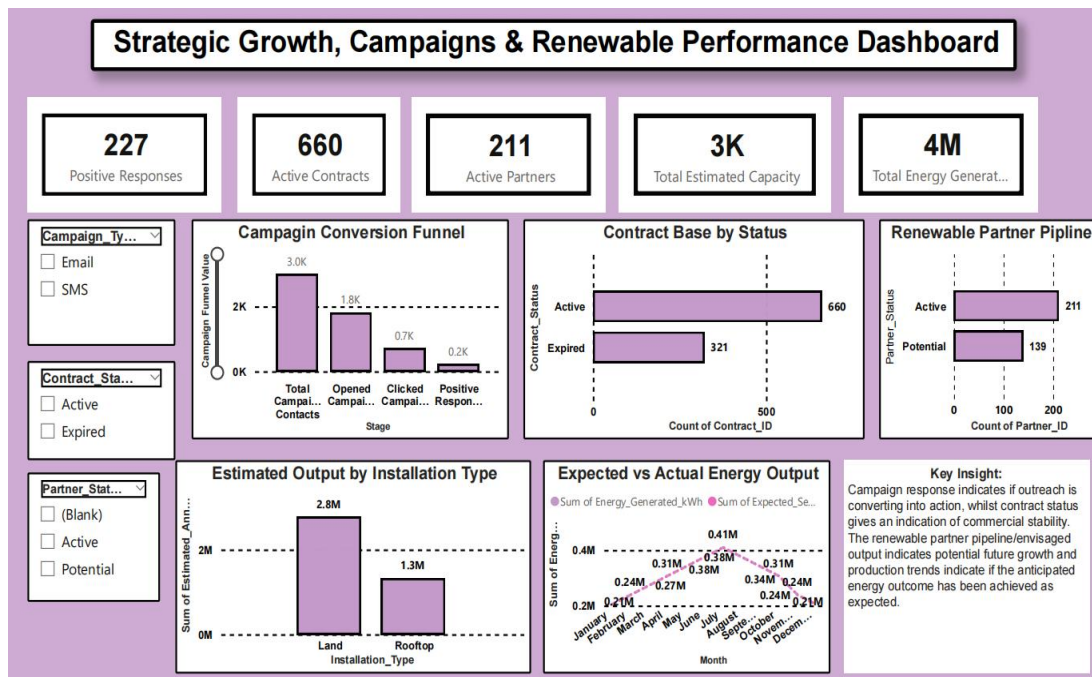


Fig 16: Strategic Growth, Campaigns & Renewable Dashboard

The Dashboard 3 shows the company growth, marketing efficiency, and the renewable energy production. This dashboard tracks this conversion process, helping the company to determine how successfully they are converting the opportunities into actual business results. The campaign conversion pipe has the substantial decrease in the conversion rates from the first point of contact to a positive final response. This shows that the company has a good deal of marketing engagement, but the actual conversion efficiency is poor. Therefore, it is necessary for the company to investigate improving their campaign targeting strategies and their follow up engagement procedures. The company currently has 660 active contracts and 211 active partners, which shows that they have a solid base from which to continue growing. Although the slow-growing partner pipeline represents the future growth opportunities that must be actively pursued. The energy production analysis includes both the expected and the actual energy produced, which identifying the variances based on the month. This shows that although the company has plans in place for energy production, execution is not consistent for many reasons such as operational or and environmental factors. Strategically, this dashboard connects marketing, operations, and the energy production, therefore, it will give management information into the current state of performance while also allowing the company to assess future growth potential. Because of the importance of the material presented in this dashboard to the decision-making process of the management team and for long-term planning, this dashboard is an essential tool for management's decision-making process.

4. Initial Response to the Implemented Solution:

The implemented CRM system and Power BI dashboards were used to determine whether they effectively address the business challenges identified in the initial analysis. These challenges included high customer churn risk, inefficient management of customer service case workloads, and a deficiency of operational visibility or the insight. The data visualisation through the BI dashboards improves the quality of the decision by providing the timely, accurate and actionable insights [5]. Although by integrating multiple data sources into a single analytical platform, as organizations can enhance forecasting, the performance monitoring, and the strategic planning [5].

A. Has the Business gap been closed?

Using the CRM system which has allowed the company to have much greater visibility of data and be able to organise data. As there is a one location where all customer data, service cases and the opportunities are now managed, thus reducing potential fragmentation while also providing the teams has increase in the accessibility of customer data. Utilising the Power BI dashboards, management can easily identify the highest-risk customers, specifically those whom are identified as the highest percentage of potential for churn (9.7%). This type of visibility was not formerly available, thus making it very difficult to develop better strategies to retain customers. By developing a profile for these identified customers, the company will be able to apply more targeted strategies to retain these customers.

B. Has the business process improved?

The use of the operational automation within the CRM system has increased the operational efficiency, particularly in supporting customer cases. An automated escalation process make sure that if a case is not resolved in 48 hours or less and is flagged as a case that should be worked on quickly as a result, there are fewer chances for unresolved high-priority cases, which leads to increased customer satisfaction. As seen in the dashboards for the services, there were many cases being escalated by the system, which shows that the automation was effectively identifying and resolving the urgent issues. Although this will help provide more visibility of escalations, as it also shows that there are still some operational problems that remain to be fixed. This may mean that, while the new management response to escalated cases may improve, as additional process optimisation must still occur.

C. Has Performance improved?

The dashboards have measurable gains in the improvement that they can track performance metrics for the organisation as a whole, such as the resolution times for resolving the customer complaints, potential for customers to leave the business, the effectiveness of campaigns and the operational output of the organisation. The ability to measure, for example, high-value billing complaints and also the level of complaints relating to services, gives management the ability to focus on the key contributing factors to the customer dissatisfaction. Similarly, the ability to access the campaign performance data allows campaign performance metrics to identify the ineffective conversion rates, thereby giving marketing teams the opportunity to create more effective campaigns. The dashboards provides the excellent analytical capabilities, however, the performance improvement depends on how effectively the management team utilises the analytical capabilities contained within the dashboards. Therefore, the dashboards provide a platform to improve the performance, but do not guarantee performance can be improved.

D. Limitations:

There are still some limitations with today's solution as the improvements have been made. The solution is based on a relatively small dataset that does not necessarily cover the complexity of the world. In addition to automated processes for the case escalation, the predictive churn prevention and the customer engagement strategies have not been fully developed. Also, because the Power BI dashboards are based on the static data, they cannot provide instantaneous operation capabilities. Therefore, these limitations shows that the current solution is the foundation for further enhancements before it is implemented on a full scale.

5. Conclusion:

By demonstrating the CRM system and the Power BI dashboards to address the main challenges faced by the organisation, the project has shown that it is possible to overcome various issues through the implementation of an integrated technology solution. The result is better organised data, more visibility into the behaviour of customers, and making the better informed business decisions. The most important element provided by this project is the ability to link operational data to analytical insights. The continued effectiveness of this CRM and Power BI solution is conditional upon the continuous improvements, quality of data, and adoption by users. If any one of these factors are not addressed, then an organisation will not benefit from the long-term value of even the most cutting-edge technology solutions.

6. Future Work:

The future improvements may be implemented, such as creating a real-time data pipeline between the Salesforce and Power BI so that live dashboards can be created. Also, the introduction of machine learning algorithms can be used to create predictions for customer churn and to recommend the targeted retention solutions for the customers. As well as, additional automation opportunities may also exist, for example automated follow-ups on high-risk customers and the intelligent routing of cases based on nature or the urgency of the issues. Finally, expanding the overall integration of the system with suppliers and the renewable business partners will provide even greater visibility of all businesses.

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